

Physics

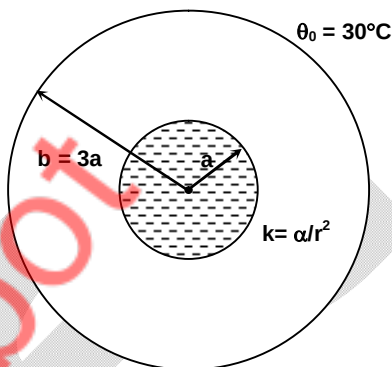
PART – I

SECTION – A

(One Options Correct Type)

This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

1. Water of mass 'm' and specific heat capacity 'S' is kept in the spherical cavity of radius 'a' of a thick spherical shell of inner and outer radii a and 3a respectively which is placed in a surrounding at temperature $\theta_0 = 30^\circ\text{C}$ as shown in the figure. Thermal conductivity of the shell varies as $k = \frac{\alpha}{r^2}$, where α is a constant. If initial temperature of water is 70°C , then the time required to decrease the temperature of water from 70°C to 50°C is



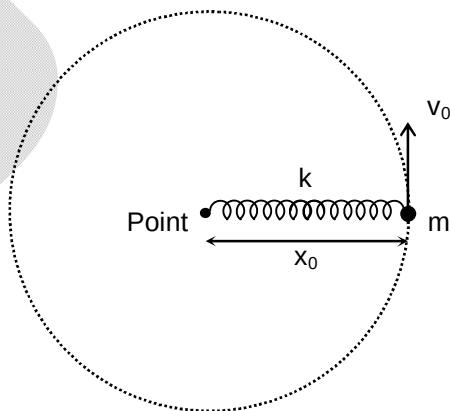
(A) $\frac{mSa\ln 2}{4\pi\alpha}$

(B) $\frac{mSa\ln 2}{2\pi\alpha}$

(C) $\frac{2mSa\ln 2}{3\pi\alpha}$

(D) $\frac{4mSa\ln 2}{3\pi\alpha}$

2. A hypothetical massless spring having negligible relaxed length and follows the relation between force F and extension x as $F = cx^{-2}$, where c is constant and x is extension in spring. Spring can only be broken if it stretched upto a very large length. A point mass is performing circular motion with help of spring as shown, with speed v_0 . The velocity needed for mass to break the spring is close to:



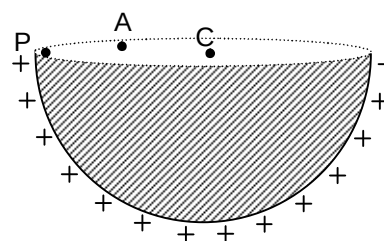
(A) $2v_0$

(B) $\sqrt{2}v_0$

(C) $3v_0$

(D) $2\sqrt{2}v_0$

3. A non conducting hemispherical shell of radius 'R' has uniform surface charge density σ . Point 'C' is center of hemispherical shell. Point 'A' and point 'P' lie on the flat surface of the shell. The distance between C and A is R/2 and between C and P is R. Choose the correct option(s).



(A) The potential at point P is $\frac{\sigma R}{2\epsilon_0}$

(B) The potential at point A is $\frac{\sigma R}{4\epsilon_0}$

(C) Work done by electrostatic force if a charge ' q_0 ' is moved from P to A is $\frac{-q_0\sigma R}{4\epsilon_0}$

(D) Work done by electrostatic force if a charge ' q_0 ' is moved from P to A is $\frac{q_0\sigma R}{4\epsilon_0}$